

A Formalization Milestone: A Machine-Checked Convergence Chain for a Periodic Cayley–Crank–Nicolson Discretization

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Abstract

We report a Lean 4.31 formalization connecting the main components of a classical convergence analysis for a periodic one-dimensional Cayley–Crank–Nicolson discretization of the linear Schrödinger equation. This formalization milestone includes the periodic centered finite-difference Laplacian, Cayley-factor invertibility and unitarity, arbitrary finite-step weighted-norm stability, explicit spatial and temporal Taylor remainder estimates, a fixed-time $O(k^2 + h^2)$ bound, and grid-family convergence. The numerical theorem is classical. The contribution is the audited Lean 4 formalization. Any claim that this is the first formalization of its exact kind remains preliminary and unchecked. The principal declarations compile without `sorry`, `admit`, custom axioms, or `native_decide`.

1 Scope and contribution

Status of this document. This is a formalization milestone. The numerical theorem is classical. The contribution is the audited Lean 4 formalization. Any “first” claim remains preliminary and unchecked.

A paper proof of numerical PDE convergence crosses analytic Taylor estimates, finite periodic indexing, complex matrix algebra, stability, and asymptotic reasoning. Machine checking forces the interfaces between these layers to be explicit. This development connects six verified stages:

1. the periodic centered finite-difference Laplacian;
2. Cayley-factor invertibility and unitarity;
3. arbitrary finite-step weighted-norm stability;
4. explicit spatial and temporal Taylor remainder estimates;
5. a fixed-time $O(k^2 + h^2)$ error estimate; and
6. a grid-family convergence result.

The work does not claim a new numerical method or analytic convergence order. Formal convergence of other finite-difference PDE schemes has been achieved in Coq [2, 3]. The present contribution is the particular audited end-to-end Lean chain and its reusable interfaces. Public searches have not established priority, and no definitive “first” claim is made.

2 Continuous problem and discrete update

Let $U : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{C}$ satisfy

$$\partial_t U(t, x) = i \partial_{xx} U(t, x), \quad U(t, x + L) = U(t, x), \quad L > 0. \quad (1)$$

For $m+1$ periodic nodes, let $h = L/(m+1)$ and let $H_h = -\Delta_h$ be the centered periodic negative Laplacian. For $k > 0$, put

$$A = I + i \frac{k}{2} H_h, \quad B = I - i \frac{k}{2} H_h, \quad Q = BA^{-1}.$$

The numerical trajectory satisfies $v^{j+1} = Qv^j$, and

$$\|w\|_h = \left(h \sum_i |w_i|^2 \right)^{1/2}.$$

3 The machine-checked chain

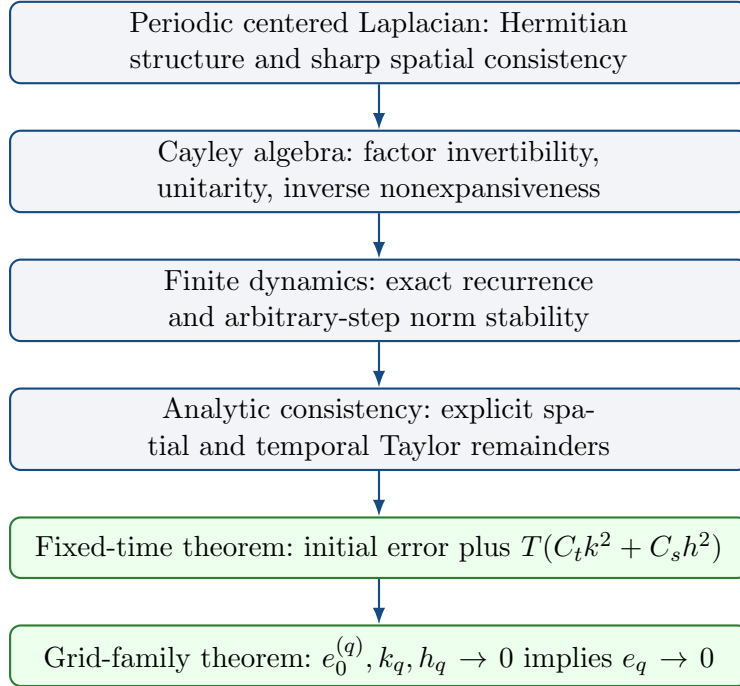


Figure 1: The verified dependency spine.

3.1 Periodic centered Laplacian

For a periodic grid vector w ,

$$(H_h w)_i = \frac{2w_i - w_{i+1} - w_{i-1}}{h^2},$$

with indices interpreted cyclically. The development proves the index identities, invariance of finite sums under shifts, the matrix action formula, Hermiticity and positive semidefiniteness of H_h , and summation by parts.

3.2 Cayley invertibility, unitarity, and finite stability

Hermiticity gives the Gram identity needed to prove that A is invertible. The formal proof constructs A^{-1} internally and verifies both multiplication identities. It then proves

$$Q^*Q = I, \quad QQ^* = I.$$

Thus Q preserves both Euclidean and weighted norms, while A^{-1} is nonexpansive in the norm used for residual accumulation. For an explicitly recursive N -step update, induction proves

$$\|Q^N w\|_h = \|w\|_h. \quad (2)$$

The dimension and the number of steps are quantified rather than computed at a single finite instance.

3.3 Explicit Taylor estimates

Assume the uniform derivative bounds

$$|\partial_x^4 U(t, x)| \leq M_s, \quad |\partial_t^3 U(t, x)| \leq M_t.$$

The two spatial integral remainders satisfy

$$|R_+(x, h)| \leq \frac{M_s h^4}{24}, \quad |R_-(x, h)| \leq \frac{M_s h^4}{24}.$$

Cancellation of the odd terms and division by h^2 gives

$$|(-\Delta_h u)(x) + u''(x)| \leq \frac{M_s}{12} h^2. \quad (3)$$

Periodic sampling and $(m+1)h = L$ yield

$$\|r_s^j\|_h \leq \sqrt{L} \frac{M_s}{12} h^2. \quad (4)$$

For the normalized Crank–Nicolson temporal defect, the verified Taylor assembly yields

$$\|r_t^j\|_h \leq \sqrt{L} \frac{5M_t}{12} k^2. \quad (5)$$

The analytic-closure module derives the needed interval statements from global C^3 time regularity rather than assuming remainder estimates externally.

4 Fixed-time error estimate

Let w_h^j sample $U(jk, \cdot)$ and set $e^j = w_h^j - v^j$. The PDE identity, averaged spatial profile, and consistency estimates give the exact residual

$$Aw_h^{j+1} = Bw_h^j + kr^j.$$

Applying A^{-1} gives

$$e^{j+1} = Qe^j + A^{-1}(kr^j), \quad \|e^{j+1}\|_h \leq \|e^j\|_h + k \|r^j\|_h.$$

Iteration and $Nk \leq T$ prove

$$\boxed{\|e^N\|_h \leq \|e^0\|_h + T \left[\sqrt{L} \frac{5M_t}{12} k^2 + \sqrt{L} \frac{M_s}{12} h^2 \right]}. \quad (6)$$

The Lean declaration is `periodic_cayley_cn_convergence_of_smooth_solution`.

5 Grid-family convergence

For a family indexed by q , if

$$e_0^{(q)} \rightarrow 0, \quad k_q \rightarrow 0, \quad h_q \rightarrow 0,$$

then the explicit right-hand side of (6) tends to zero. A squeeze argument gives convergence of the numerical error. Exact initialization is a specialized corollary.

6 Lean artifact and assurance

Module	Role
PeriodicCayleyCN-AnalyticClosureV1	Smooth PDE solution to fixed-time estimate
PeriodicCayleyCN-ConstantModeExampleV1	Exact zero-error constant-mode check
PeriodicCayleyCN-GridFamilyConvergenceV1	Vanishing-mesh convergence consequences

The production sources were built with Lean 4.31.0. Clean-runtime replay, direct source checks, theorem-signature inspection, and axiom audits were preserved. The headline declarations depend only on `propext`, `Classical.choice`, and `Quot.sound`. The relevant new sources contain no `sorry`, `admit`, custom `axiom`, or `native_decide`.

Kernel acceptance alone does not ensure that the encoded statement is the intended one [4]. Accordingly, the project preserves the informal theorem, explicit constants, dependency map, anti-placeholder checks, and a prior-art/reviewer packet. Any priority assessment remains preliminary and unchecked.

7 Limitations and conclusion

The theorem concerns a smooth solution of the linear one-dimensional periodic equation. It is not a nonlinear well-posedness theorem, a long-time dispersive analysis, a floating-point program certificate, or an infinite-dimensional operator-convergence theorem.

This formalization milestone is a connected, machine-checked account of how periodic finite differences, Cayley algebra, stability, Taylor consistency, finite-time accumulation, and asymptotic convergence fit together. It provides an auditable certificate for the stated result and a foundation for further formal numerical-PDE developments in Lean.

References

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- [3] M. Tekriwal, K. Duraisamy, and J.-B. Jeannin, *A formal proof of the Lax equivalence theorem for finite difference schemes*, 2021. <https://arxiv.org/abs/2103.13534>
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